Semester: 5th
Subject Name:- CI & Code:- CS30011
Branch (s): - CSE, IT, CSCE

AUTUMN **MAKE UP MID SEMESTER EXAMINATION-2024**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Computational Intelligence
[CS30011]

Time: 1 1/2 Hours **Full Mark: 20**

Answer all the questions.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 X 5]

a) Consider an MLP neural network with five nodes in the input layer, seven nodes in the hidden layer and nine nodes in the output layer. Calculate the total number of weights and biases in the network.

b) What is the main difference between Perceptron and Adaline?

c) Explain the purpose of mutation in GA.

d) Calculate $A_{0.4}$ for the fuzzy set A = { 0.1/x1 + 0.3/x2 + 0.6/x3 + 0.9/x4 + 0.7/x5 + 0.4/x6 + 0.2/x7}

e) Define the Sugeno's complement operator.

2.

a) Design a single-unit McCulloch-Pitts artificial neuron with two inputs that can implement the following logic gate function OP. [2.5]

X_1	X_2	X_1 OP X_2
0	0	1
0	1	0
1	0	1
1	1	1

b) A 2-input neuron, single output ANN has weights values [3 2],bias=1.2.It is given an input $[-5\ 6]^T$. Find the output for the following transfer functions: [2.5]
(i) Symmetrical hard limit (ii) Saturating Linear (iii) Hyperbolic tangent sigmoidal

3. [5]

Mention the basic steps of GA preferably through a flow chart.
Apply the steps of GA to maximize the function:

$y = 31x - x^2$ where x is an integer, $0 \leq x \leq 31$

Assume rate of crossover to be 100% and rate of mutation to be 0%.
Show at least two generations starting from initial population. (Initial population is not considered as a generation.)

4.

a) Define composite and non-composite MFs. [2.5]

Find the Cartesian product and Cartesian co-product for the given fuzzy sets –

A = 0.1/x1 + 0.4/x2 + 0.8/x3 + 1/x4 + 0.5/x5

B= 0.2/y1 + 0.3/y2 + 0.6/y3 + 0.9/y4 + 0.7/y5

b) Define the following terms with examples in the context of fuzzy set theory – [2.5]

(i) Support (ii) Crossover point (iii) Bandwidth

(iv) Convexity (v) Symmetry

*** Best of Luck ***